

Impact of HVDC Transmission System Topology on Multiterminal DC Network Faults

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Abstract—This paper compares how a dc fault affects a multiterminal dc (MTdc) network depending on the HVDC transmission system topology. To this end, a six-step methodology is proposed for the selection of the necessary dc fault protection measures. The network consists of four voltage-source converters connected radially. The converters natural fault response to a dc fault for the different topologies is studied using dynamic simulation models. For clearing of the dc faults, four different dc breaker technologies are compared based on their fault interruption time, together with a current direction fault detection method. If necessary, the converters are reinforced with limiting reactors to decrease the peak value and rate of rise of the fault currents providing sufficient time for the breakers to isolate the fault without interrupting the MTdc network operation. The study shows that the symmetric monopolar topology is least affected by dc contingencies. Considering bipolar topologies, the bipolar with metallic return exhibits better fault response compared to the one with ground return. Topologies with ground or metallic return require full semiconductor or hybrid breakers with reactors to successfully isolate a dc fault.

Index Terms—Control systems, fault currents, HVDC circuit breakers, HVDC converters, HVDC transmission, multiterminal networks, voltage-source converters (VSCs).

I. INTRODUCTION

THE BEST sites for the exploitation of renewable resources are usually remotely located from demand centers since these sources have a low power-area ratio. Since space is scarce and costly in densely populated urban regions, long-distance power transmission systems will be vital for higher exploitation of available global energy resources and an increase in energy trade.

Especially in cases where electricity needs to be transported via long lines, high-voltage direct current (HVdc) transmission has several advantages over its high-voltage alternate current (HVAC) counterpart, for example, greater power per conductor, fewer problems with resonances, and transmission distance that is not limited by stability. However, out of more than 140 HVDC

projects worldwide, only two are multiterminal networks, and they have more than three converter terminals connected via a dc network [1].

Nevertheless, four key aspects need to be thoroughly analyzed before multiterminal dc networks can be successfully developed: system integration, dynamic behavior, power-flow control, and fault behavior.

System integration aspects of multiterminal dc networks have been tackled in [2]–[5]. The dynamic behavior of voltage-source converters (VSCs), including multilevel topologies and multiterminal dc networks, has been extensively analyzed in the literature in [6]–[8]. The control of the power flow in MTDC networks has also been thoroughly analyzed in [7] and [9]–[11].

The response of MTDC networks to contingencies, specially on the dc side of the transmission system, will be vital for its successful development. After a dc fault, the system will most likely have only a few milliseconds to clear the fault before high dc currents pose a threat to the MTDC network [12].

Therefore, several efforts have concentrated on research and development of HVDC switch breakers [13], [14]. Previous research on the protection of high-voltage MTDC networks have also focused on how a dc fault develops [15]–[17]; dc fault detection methods [18], [19]; and protection systems [20]–[22].

The main contribution of this paper is the comparison of different HVDC transmission system topologies under dc fault cases and the analysis of different current limiting reactors and the dc switch breakers' impact on the developing dc fault currents.

This paper is structured as follows: Section II briefly introduces the most common HVDC transmission system topologies. In Section III, the available dc breaker technologies for the protection of MTDC networks are discussed. The case study is established in Section IV, and a methodology is presented to select dc breaker technology and establish whether dc protection reactors are necessary. In Section V, representative results are presented. These were obtained via a numerical dynamic simulation model, which allows the performance comparison of different HVDC topologies. Finally, in Section VI, in accordance with the obtained results, conclusions and recommendations are drawn.

II. HVDC TOPOLOGIES

Independently from the converter technology, there are two main topologies for HVDC transmission systems: monopolar and bipolar [23]. Table I displays the topologies analyzed in this paper; all of which can be employed to form MTDC networks [23], [24]. Fig. 1 depicts a single-phase representation of

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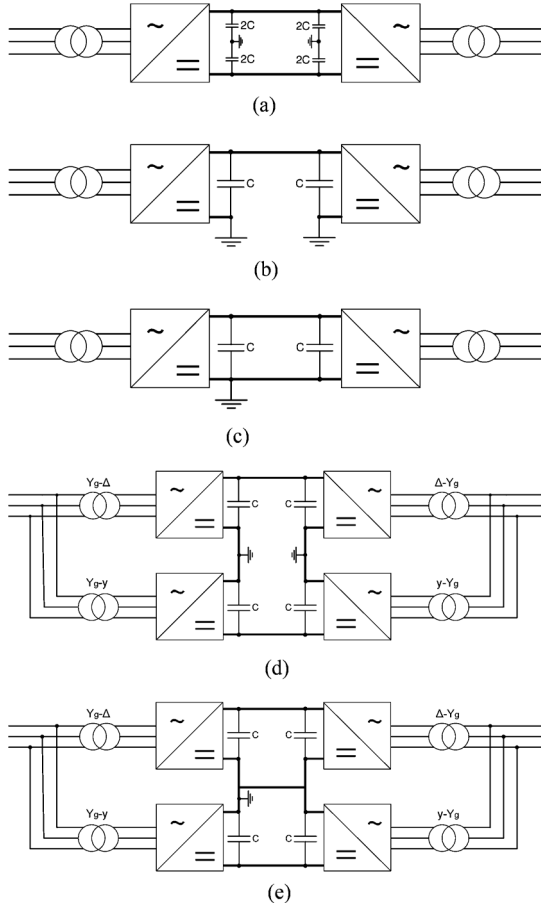


Fig. 1. HVDC transmission system topologies. (a) Symmetric monopole. (b) Asymmetric monopole with ground return. (c) Asymmetric monopole with metallic return. (d) Bipolar with ground return. (e) Bipolar with metallic return.

TABLE I
ANALYZED HVDC TRANSMISSION TOPOLOGIES

	Topology			
	A. Monopolar		B. Bipolar	
Return Path	(a)	Symmetric	(d)	Ground Return
	(b)	Assymmetric Ground	(e)	Metallic Return
	(c)	Assymmetric Metallic		

two HVdc converter stations connected to three-phase ac systems via a transformer. The converter configuration is abstracted as the main HVDC transmission system topologies are distinguished by their dc circuit schemes.

A. Monopolar HVDC Topologies

As the name suggests, this configuration has only one pole, mostly of negative polarity to reduce corona effects [23]. Monopolar topologies as well as bipolar topologies can be further classified according to the return path used.

a) *Symmetric Monopole*: This configuration either uses no grounding at the dc circuit side or—in VSC–HVdc systems—the dc-link capacitors are grounded at their middle point. Due to its symmetry, this is the only topology where the ac transformer is not subjected to dc voltage stresses, which simplifies its design. In addition, it is the only topology where

in the case of a pole-to-ground fault, the dc side is not fed by ac-side currents. However, it has higher costs since it requires two fully insulated conductors.

b) *Asymmetric Monopole With Ground Return*: This topology is advantageous from a cost perspective due to the presence of only one fully insulated conductor and the possibility of expansion to a bipolar if necessary. However, it requires permission for the installation of electrodes and continuous ground current, which can raise environmental concerns.

c) *Asymmetric Monopole With Metallic Return*: This topology has no ground currents, since the return is made via the metallic conductor, which does not require high-voltage insulation. This topology can be easily expanded to a bipolar.

All monopolar HVDC topologies lack redundancy as, in the case of a dc fault, all terminals are affected by the high fault currents. Unless selective protection methods that are able to isolate the faulty HVDC line in time are implemented, the MTDC network has to be de-energized before operation is restored.

B. Bipolar HVDC Topologies

Bipolar configuration employs at least two converters at each network terminal, which makes it a more costly alternative than the monopolar one. However, bipolar topologies can achieve twice the power rating of monopolar topologies.

Each converter is connected to the ac grid via two different transformer groups, or by a transformer with two secondary windings. Usually one pole uses a Yg–Δ transformer configuration whereas the other uses a Yg–Y configuration, to reduce the ac-side harmonic distortion [1].

As long as the dc side of the transmission system is grounded, the transformers secondary windings need to be designed for high direct voltage stresses, namely, half the direct voltage nominal value. Hence, special attention has to be paid to their insulation. The main advantage of the bipolar configuration is its redundancy, which can be more than half the total transmission capacity if poles can be overloaded whenever one converter suffers a fault. However, there are disadvantages for each of the available bipolar topologies.

a) *Bipole With Ground Return*: In addition to higher costs, when compared to respective monopolar configurations, the bipolar configuration also raises environmental concerns due to ground currents in cases where both poles are not balanced, for example, pole outages or maintenance periods. Even though not necessary, grounding is often provided for the bipolar configuration. If there is no return path for the current, the entire HVDC transmission system will be made unavailable during faults.

b) *Bipole With Metallic Return*: Compared to the bipolar with ground return, this topology needs an extra low-voltage insulated neutral conductor. If injecting current in the ground is not possible, there is the possibility to use a fully insulated metallic return and use it as a spare cable in case of a fault, providing a more flexible albeit even more expensive solution.

III. BREAKER TECHNOLOGIES

Although the line-commutated current-source converter (LCC–HVdc) remains the most mature HVDC technology, its

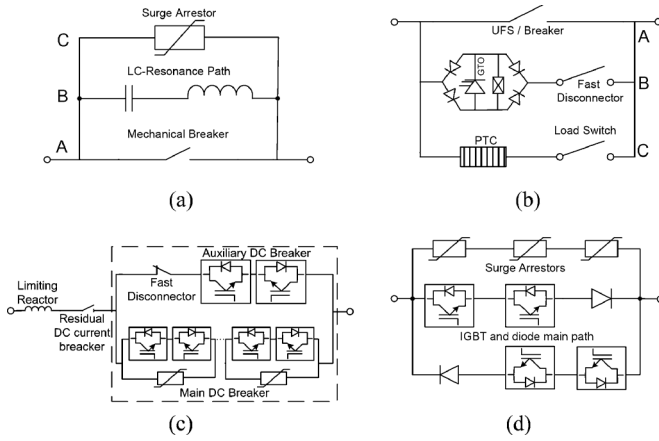


Fig. 2. HVDC breaker technologies. (a) Resonant breaker. (b) Hybrid II breaker. (c) Hybrid I breaker. (d) Solid-state breaker.

inherent low controllability as well as difficulties in forming MTDC networks, have driven researchers and the industry toward the application of VSC for transmission purposes.

In a VSC-HVDC system, which uses fully controllable switches, power-flow reversal only involves changing the converter current direction, while the direct voltage remains constant, making it easier to develop MTDC networks. However, the use of insulated-gate bipolar transistor (IGBT) valves is, at the same time, a main drawback. In case of a dc fault, contrary to thyristor valves, IGBT valves cannot block the fault current due to the current path via the anti-parallel diodes which makes converters prone to damage. This issue becomes more challenging in an MTDC network where the costs of power transfer loss are high.

As a result, dc breakers are required to isolate the fault and resume normal operation without interruptions. Next, an overview of the most common dc breaker technologies—shown in Fig. 2—is provided.

A. Resonance Breakers

Resonance breakers—composed of three current paths—are based on ac gas-insulated substation (GIS) breakers. During normal operation, the current flows through a low-loss switch, which opens as soon as a fault is detected, creating an arc. Since the arcing voltage is high, an RLC path is needed to limit the commutation voltage.

The energy-storage elements create a resonant current through the main path switch, which can interrupt the current when this crosses zero. The surge arresters limit the maximum voltage across the resonant circuit capacitor and the current is commutated to the energy absorbers which bring it down to zero [14], [25], [26].

The main advantages of this breaker technology are its low cost and low onstate losses. However, they are significantly slower than the other investigated technologies [26].

B. Hybrid Fault-Current Limiting Breaker (Hybrid II)

Hybrid breakers combine the use of fast semiconductor switches with current limiters, or mechanical switches, and usually consist of three parallel paths [see Fig. 2(b)] [27].

The first path, consisting of either of a mechanical breaker or an ultrafast switch (UFS), commutates the current to the second path. The second path includes gate turnoff thyristors (GTOs) and a diode bridge to accommodate bidirectional fault currents. As soon as the current crosses zero on the second path, a fast disconnecting switch is opened to isolate it. Finally, the current is commutated to the third path, which acts as a current limiter [27]. This path has a high positive temperature coefficient (PTC) resistor and a load switch, which can interrupt current at the first zero crossing.

A mechanical switch is inserted in series to meet the voltage-withstand requirements of the dc breaker [28], resulting though in increased interruption times (≈ 20 ms) [26].

C. Hybrid Solid-State Breakers (Hybrid I)

This dc breaker aims at a compromise between switching speed and onstate losses by using two current paths: one for conduction and one for interruption (main breaker path).

When a fault occurs, the auxiliary dc breaker commutates the current to the main breaker path and a fast mechanical disconnector opens. Since the auxiliary breaker only needs to commutate the current to the main breaker path, it can have lower voltage and current ratings so that its on-state losses are minimized during normal operation.

Because of the mechanical disconnector, the interruption time is higher than for full solid-state breakers. Nevertheless, opening times of 2 ms are possible if several mechanical switches are connected in series, as in the modular hybrid IGBT dc breaker proposed by ABB [13].

D. Solid-State Breakers

In solid-state breakers, a large number of semiconductor switches are connected in parallel and series to withstand the maximum breaking current and voltage. Several topologies using solid-state switches have been proposed [26], [27], [29], but generally the breaker consists of three paths.

There are two conduction paths, for bidirectional power flow, in which the number of switches depends on the breaker voltage rating. A diode is used to prevent currents to flow through the switches' antiparallel diodes when these are blocked. As soon as a fault is detected, the switches are blocked and the current commutates to the parallel-connected varistors. The voltage and current across the main interruption path are suppressed by the surge arresters [29].

Since switching speed is essential for MTDC networks, IGBTs and gate-commutated thyristors (GCTs) are preferred in solid-state breakers, and the total interruption time is kept around 1 ms [26], [27]. However, the use of semiconductors in series introduces significant onstate losses [27]. Hitherto, solid-state breakers have mainly been proposed for LVDC and MVDC applications [26].

A summary of the most important parameters of the dc breakers technologies discussed is given in Table II [26].

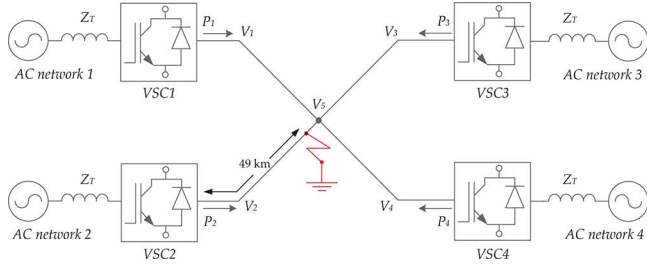


Fig. 3. Layout of the analyzed radial MTDC network with four terminals.

TABLE II
DC BREAKER TECHNOLOGIES COMPARISON

	Solid State	Hybrid I	Hybrid II	Resonance
Commutation [ms]	switch: 0.1	switch: ≤ 0.2 ; breaker: ≈ 0.25	switch: ≈ 0.1 ; breaker: ≤ 20 ; UFS: 1-5	breaker: ≤ 20 ; resonance: ≈ 30
Energy absorption [ms]	≈ 1	≈ 1	-	-
Interruption time [ms]	< 1	2-5	5-30	< 60
Interruption time simulated (t_b) [ms]	1	2	30	60
Max. voltage [kV]	≤ 800	tested: 120; expected: 320	AC-CB: ≥ 500 ; UFS: ≤ 12	available: ≤ 550
Max. current [kA]	≤ 5	tested: 9; expected: 16	estimated: 6 - 12	tested: 4
Losses [% converter ratings]	GCT $\approx 30\%$ GTO $\approx 40\%$	$\leq 1\%$	negligible	negligible

IV. CASE STUDY AND METHODOLOGY

A. Case Study

The aim of this paper is to compare the different HVDC topologies for the development of MTDC networks and to provide a methodology for selecting protection measures against dc faults. The layout of the MTDC network used in this paper is shown in Fig. 3. A four-terminal grid with a radial configuration was selected since most likely early MTDC networks will have a limited number of terminals. Moreover, a radial network can be considered more challenging than a meshed one since, in case of a dc fault, a converter has to be completely isolated from the network. The main MTDC network parameters that are used are shown in Table III.

To analyze the dc fault behavior of the MTdc network, a dynamic simulation model was developed using the SimPower-System toolbox of Matlab/Simulink. Since it is difficult to analyze the VSC response to a dc fault using an averaged model, all simulations were performed using a model which includes the converter switching behavior.

B. Methodology

The workflow of the proposed methodology is presented in Fig. 4. Each step of the methodology is explained next.

1) *HVdc Topology*: The starting point is the selection of a topology for the MTDC network. In this paper, all topologies shown in Table I and displayed in Fig. 1 are compared through means of a dynamic simulation model.

2) *DC Fault Selection*: Once the dynamic simulation model is built, a dc fault type and location needs to be selected. There are two main types of dc faults, which can occur in an MTDC grid: pole-to-ground fault and line-to-line fault, or short-circuit

TABLE III
MTDC NETWORK PARAMETERS

Grid parameters	Unit	Value
VSC rated power (S_{VSC})	MVA	1200
MTdc voltage (V_{dc})	kV	± 320
Rated current (I_{dc})	kA	2.062
Cable Resistance (r)	Ω/km	0.0195
Cable Inductance (l)	mH/km	20
Cable Capacitance (c)	nF/km	220
Cable distance (d1/d2/d3/d4)	km	50
DC link capacitance (C)	μF	100
Phase Reactor Impedance (Z_{pr})	pu	$0.001+j0.1$
AC Network HV side (E_h)	kV	380
AC Network LV side (E_l)	kV	150
Transformer rated power (S_T)	MVA	1200
Transformer impedance (Z_T)	pu	$0.001+j0.1$

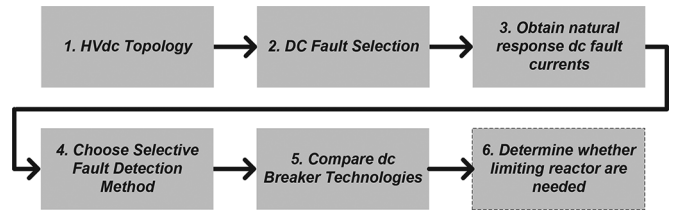


Fig. 4. Workflow of the methodology used in the topologies study.

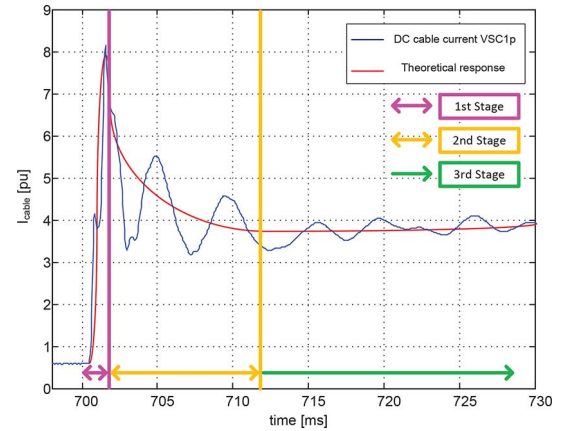


Fig. 5. Theoretical and simulated fault current response of a VSC in case of a pole-to-ground fault.

fault. If underground or submarine cables are used, line-to-line faults are very rare when compared to pole-to-ground faults; hence, the latter was selected [30]. All pole-to-ground faults were simulated with a ground resistance of 7Ω , corresponding to wet loamy sand ground type [31].

In addition to selecting the fault type, the fault location also needs to be determined. Three different fault locations – 1, 25, and 49 km (measured from VSC2) were analyzed. It was found that the closer the fault is to the middle point of the MTDC network, the less time it takes to reach the fault current peak and the higher is the peak value. Fig. 6 shows the current directions in the MTDC network in case of a fault on the dc line leading to VSC2.

3) *Obtain Natural Fault Current Response*: Generally, dc faults are developed in three stages. In the first stage, VSC capacitors and the line energy-storage elements start discharging

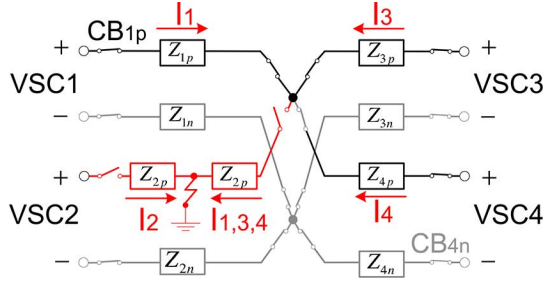


Fig. 6. Current direction in case of a positive pole dc fault on the VSC2 line. Z_{ik} : Impedance of line i at positive “p” pole or negative “n” pole. CB_{ik} : DC circuit breaker of line i at positive “p” pole or negative “n” pole.

through the fault. The second stage starts as soon as the overcurrent protection of the converter is triggered. At this moment, the converter switches are blocked for their protection and the VSC acts as a diode bridge rectifier. Finally, the MTDC network currents reach a new steady state after the transient response is over. Fig. 5 shows the theoretical and simulated fault current response of a VSC in case of a pole-to-ground fault, resulting from the developed dynamic models. The equations used for the calculation of the theoretical fault current response were obtained from [15].

It is known that circa 5 ms is the maximum time VSC valves can withstand the maximum short-circuit current [32]. However, in our natural fault response analysis, the converters are assumed to remain connected even after this period. It has to be noted that the overcurrent protection is triggered as soon as the IGBT valve ac current exceeds 2 p.u. [33].

4) *Determined Selective Fault Detection Method*: In addition to the overcurrent protection implemented in the VSC-HVDC terminals, to protect the converter valves, a selective fault detection method has to be determined to distinguish the faulty line from the rest of the network.

The main objective of a fault detection scheme is to reliably detect the fault as fast as possible. In this study, the current direction fault detection method was selected. This method works as follows: in practice, current measurements are being made at the dc breakers, namely, at both ends of each line—the VSC terminal and the central node of the network—and the signals are transmitted to the controller of the converter that lies at the line end, where they are processed. The controller is responsible for generating the signals that block the dc breakers and the valves of the converter. For the signal transmission, fiber optics were considered and the transmission time was taken into account in the converter's controller delay time t_{ctr} . During normal operation, the currents at both line ends have the same direction. However, in case of a dc fault, the fault point is fed with current from both line sides, first by the discharging dc lines and dc-link capacitors. As a result, only the faulty line experiences currents with opposite directions at each end (see Fig. 6). As soon as this occurs, a fault signal is generated by triggering the line dc breakers and blocking only the converters for which the dc breaker at their terminals needs to open. To avoid any false signal resulting from oscillations in the MTDC grid, the fault signal is generated only if the opposite currents at the line ends both exceed a current threshold of 2 p.u.

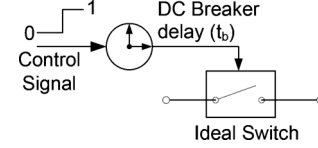


Fig. 7. DC breaker model.

It has to be noted that in case the central node of the network lies offshore, dc breakers will certainly need to be placed on an offshore platform, which may also include other components, such as control-related hardware and power-electronic converters.

5) *Compare DC Breaker Technology*: In this study, focus is given on the time response of the MTDC network to the fault. The breaker technology was assumed not to influence the MTDC network dynamic response as, under normal operation, it does not change the characteristics of the dc lines. Therefore, the dc breaker was modelled as an ideal switch with a time delay. The model is shown in Fig. 7 and the time delays—interruption times (t_b)—used for the different simulated dc breakers are summarized in Table II.

Moreover, all breaker models were included in the simulations, albeit the hybrid solid-state breaker with mechanical disconnect (Hybrid I) is, hitherto, only available for voltages lower than those simulated (± 320 kV).

6) *Define Whether Limiting Reactors Are Needed*: The total time necessary to realize and isolate a fault t_{fault} before any converter station becomes affected depends on three basic factors: the time it takes for the selective fault detection method to signal the fault t_d , the dc breaker interruption time t_b (given in Table II), and the line transport delay t_l , i.e.,

$$t_{fault} = t_d + t_b + t_l. \quad (1)$$

On the other hand, the time it takes for the overcurrent protection of a VSC to be triggered, t_{VSC} is given as

$$t_{VSC} = t_l + t_{ctr} + t_{(I_0 \rightarrow 2pu)} \quad (2)$$

where t_l is the line transport delay from the fault point, t_{ctr} is the converter's controller delay, and $t_{(I_0 \rightarrow 2pu)}$ is the time the converter current needs to surpass the overcurrent threshold value.

Finally, the line transport delay is calculated as

$$t_l = \frac{d}{v} = d\sqrt{lc} \quad (3)$$

where d is the line length and l, c are the line parameters unit-length inductance and capacitance. When calculating the transport delay, the limiting reactor inductance also needs to be taken into consideration.

To ensure system protection and resume normal operation at the postfault state, without affecting MTDC network operation, the time in which the VSC-HVDC terminals experience overcurrents has to be higher than the total fault detection and interruption time

$$t_{fault} < t_{VSC}. \quad (4)$$

TABLE IV
ORDER OF EVENTS IN BIPOLAR CONVERTERS FOR SIMULATION OF THE
LINE-TO-GROUND FAULT

Time (s)	P_{VSC2} [pu]	P_{VSC3} [pu]	P_{VSC4} [pu]
0.5	-0.3	0.3	0.4
0.7	fault	0.3	0.4

Therefore, increasing the limiting reactor size has two main beneficial effects: it delays the overcurrent protection triggering and reduces the fault current peak value. However, the sizing of the limiting reactor will be given as a compromise between dc fault current requirements and costs. For air-core reactors, the cost (C) and mass (M) are given by the following basic scaling rule [34]:

$$C, M \propto I\sqrt{L} \quad (5)$$

where I is the coil nominal current and L is the reactor inductance.

Based on the limiting reactor value, new natural fault current responses can be obtained. Therefore, it is possible to compare the new peak current values, and the time they occur, with the MTDC network response without any reactors. If the value of the peak current is higher than the rated current of the dc breaker, then another breaker technology must be chosen or the limiting reactor must be increased. Analogously, if $t_{VSC} < t_{\text{fault}}$, a higher value of limiting reactor must be selected.

V. RESULTS

As pole-to-ground faults are investigated, dynamic simulations results have shown that in the bipolar configuration, the dc fault did not influence the negative pole converters. Before, during, and after the fault, the negative pole converters kept operation independently from the positive pole converters. The same analysis has shown that the fault response of an asymmetric monopolar with ground return can safely be assumed analogous to that of the bipolar topology with ground return. The same reasoning applies to the asymmetric monopolar with metallic return topology, where the fault response is equivalent to that of the bipolar topology with metallic return. The only difference between the bipolar and the asymmetric monopolar topologies, is the inability of the latter to continue normal operation during the pole-to-ground fault.

Hence, the results are presented for the symmetric monopolar topology [topology (a) in Fig. 1], the ground return topologies [topologies (b) and (d)] and the metallic return topologies [topologies (c) and (e)].

In all simulations, the current was monitored at the converters, dc capacitors, dc lines, and at the fault point. The total simulation time was 1 s and the fault was applied on MTDC line 2, 49 km away from VSC2, at $t = 700$ ms. The MTDC network voltage is controlled by the VSC1 terminal whereas all other terminals operate in current regulation mode controlling their active power. The order of events in the MTDC network is shown in Table IV. It is worth noting that the VSC stations power references were not changed before and after the fault.

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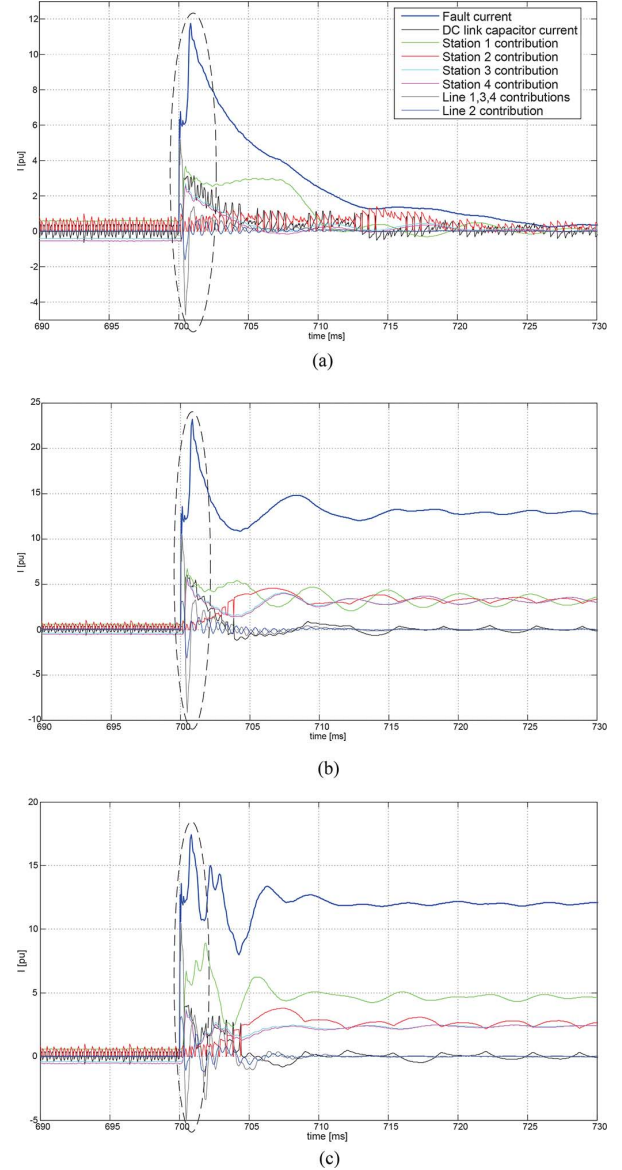


Fig. 8. Current response to a pole-to-ground on the MTDC line 2. (a) Symmetric monopole. (b) Ground return: bipolar and asymmetric monopolar. (c) Metallic return: bipolar and asymmetric monopolar.

A. Natural DC Fault Responses

After selecting the HVDC topology and applying the dc fault, the next step is to obtain the natural dc fault response of the different HVDC topologies (see Fig. 4). Fig. 8 displays the total fault current response and the contributions from the different components, when no action is taken to interrupt the fault. The important aspects of Fig. 8 are the peak dc fault currents, the relative difference between the peak currents of the different components, and the time at which the new postfault steady state is reached.

As discussed in Section IV, the fault currents develop in three phases—two transient and one steady-state phase. The initial peak of the fault current is marked with an ellipse in Fig. 8 and it can be seen that the transient period lasts for circa 15 to 20 ms.

In all topologies, the fault current is initially fed only from the discharging of the dc lines and the dc-link capacitors. After circa

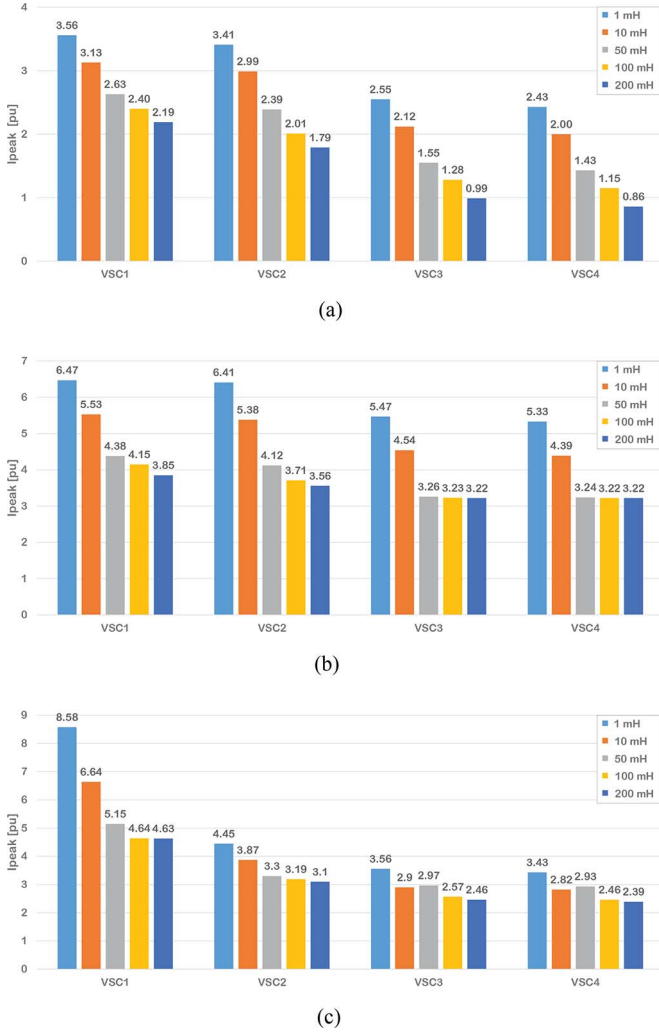


Fig. 9. Peak fault dc currents as a function of the limiting reactor size. (a) Symmetric monopole. (b) Ground return: bipolar and asymmetric monopolar. (c) Metallic return: bipolar and asymmetric monopolar.

1.0 ms, namely, the line traveling time, the VSC terminals start feeding the fault and VSC1, which is not actively controlling its active power, is the terminal that contributes first and the most.

The topologies with ground return have the worst performance compared to other topologies. The total fault current reached a higher peak, since there is no return path impedance, and the fault affected the converters within less than 4 ms, up to 2 ms faster than in the case of metallic return topologies. Except for the VSC1 converter, the converters' currents reached higher peaks in the steady-state phase, namely, 4 p.u., which can produce permanent damage.

In the topologies with metallic return, the peak fault current reached a value of 18 p.u. during the transient period. The grounding point of the metallic conductor was identified as a crucial issue regarding the dc fault response of these topologies, since it significantly influences the current contributions to the dc fault during steady state. It also affects the time required for each VSC to experience an overcurrent, which is important for designing selective fault detection methods.

Moreover, from all investigated topologies, the symmetric monopole appears to have the best response to pole-to-ground

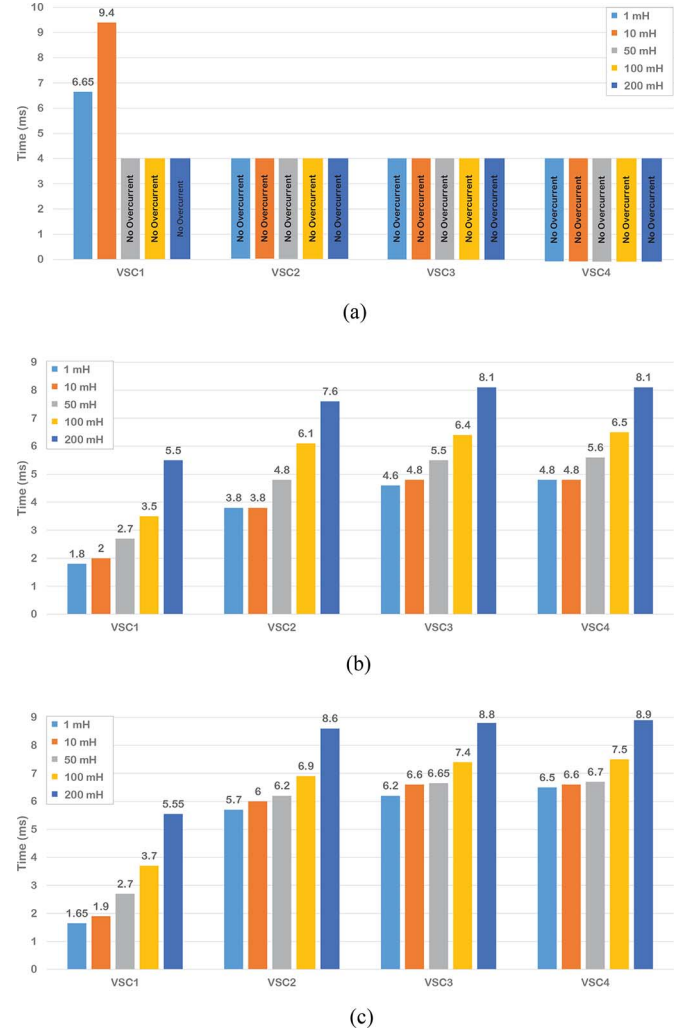


Fig. 10. AC overcurrent protection triggering time as a function of the limiting reactor size. (a) Symmetric monopole. (b) Ground return: bipolar and asymmetric monopolar. (c) Metallic return: bipolar and asymmetric monopolar.

faults. During the simulation, only VSC1—which is responsible for controlling the MTdc network voltage level—experienced overcurrents. In addition, as soon as the dc lines and positive dc-link capacitors discharge, the negative dc-link capacitors charge to hold the entire dc-link voltage, and the fault current decays to zero in the steady-state phase, approximately 25 to 30 ms after the fault occurrence.

B. Fault Current Peak Value

The graphics in Fig. 8 show that the converters have a limited time to react to the fault currents and that the currents' peaks are excessive. Hence, limiting reactors are needed to increase the time before the converters go into overcurrent protection and to lower the fault current peaks. The values of the fault dc current peak for each VSC-HVDC terminal cable, that is, after the converter output dc capacitor, are shown in Fig. 9.

The symmetric monopole topology displays the smallest overcurrents once limiting reactors are installed. Between the ground and metallic return topologies, the topologies using metallic return have the advantage of additional cable

impedance, which further limits the fault currents. Thus, regarding the value of the peak current, it is more advantageous to build the MTDC network using a metallic return.

C. Overcurrent Protection Time

The last step in the proposed methodology is to analyze whether the time in which the VSC-HVDC terminals experience an ac overcurrent is lower than the time needed for a particular dc breaker topology to take action and interrupt the fault current. The ac overcurrent protection times, as a function of the analyzed limiting reactor sizes, are displayed in Fig. 10.

For the symmetric monopole topology, only VSC1 suffers from ac overcurrents higher than 2 p.u. if limiting reactors less than 10 mH are used. If a 50 mH or higher reactor is employed at VSC1, the converters do not experience any ac overcurrents.

In this study, the line transport delay is equal to 1.1 ms and the current direction method needs 0.6 ms to detect the fault. Therefore, the total dc fault interruption time is 2.7 ms if a solid-state dc breaker is used, and 3.7 ms for the Hybrid I dc breaker and higher than 30 ms for the other two technologies.

Taking into account (4) and the times shown in Fig. 10, the passive resonant and the Hybrid II dc breakers cannot be used to isolate the fault in case of MTDC network topologies with ground or metallic return. If the Hybrid I dc breakers are used, limiting reactors higher than 100 mH are needed to enable successful fault interruption. In case of solid-state dc breakers, limiting reactors of over 50 mH are necessary.

VI. CONCLUSION

A methodology has been proposed to compare different VSC-HVDC topologies with regard to faults on MTDC networks. The proposed methodology consists of six steps which are carried out based on results from a dynamic simulation model of the complete system.

Among all analyzed topologies, the symmetric monopolar has the best fault response, especially in combination with at least 50-mH reactors. If the power to be transmitted in the dc network requires the use of bipolar topologies, the bipolar topology with metallic return, although it has higher capital installation costs, has superior performance with regard to dc faults than the one with ground return.

In conclusion, a successful fault isolation in topologies with ground or metallic return requires solid-state breakers or Hybrid I breakers with reactors higher than 50 mH or 100 mH, respectively. Since the breakers' interruption time is somewhat uncertain, sensitivity analysis can be performed using the proposed methodology. The other two breaker technologies—the Hybrid II and the resonant breakers—are not yet fast enough to handle dc contingencies for these grid topologies.

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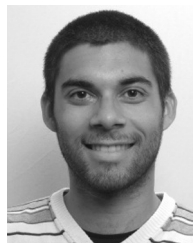
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